Benjamin Reynolds

Research Systems Engineer | Scientific Software | Automation | Hardware Integration

Denver, CO | 303-547-2170 | Benjamin.g.reynolds@ucdenver.edu
linkedin.com/in/benjamin-reynolds | github.com/bengreynolds

PROFESSIONAL SUMMARY

Research engineer building scientific software, automated behavioral systems, acquisition and control workflows, and hardware-integrated tools. Works across design, deployment, validation, documentation, and support, with an emphasis on reliable operation, reproducible data, and practical handoffs.

EXPERIENCE

Intermediate Research Engineer
Nov 2023 - Present

University of Colorado Anschutz - Department of Physiology & Biophysics

- Design, build, and optimize automated behavioral systems, desktop control interfaces, custom data pipelines, embedded control software, and analysis tools.
- Architect NWB-formatted data pipelines to improve long-term accessibility, interoperability, and reproducibility.
- Lead software and hardware upgrade cycles with vendor coordination, testing gates, deployment validation, and recovery planning.
- Maintain and standardize department-wide behavioral rigs through calibration, repair, configuration control, inventory, preventive maintenance, documentation, SOPs, and user training.

Lead Researcher & Project Assistant
May 2023 - Oct 2023

CU Denver CIDE

- Led a systematic literature review on assistive technology for people with mild cognitive impairments.
- Designed three mixed-methods studies evaluating audio-visual prompting and iOS accessibility features, then completed analysis and formal reporting.

Web Developer & Database Framework Architect
May 2022 - Sep 2022

Peak Leadership Frameworks

- Built secure web applications and modular database frameworks with SQL Server, Caspio, HTML, JavaScript, authentication, and encryption.

Senior Design Project - Lead Engineer & Project Manager
Aug 2022 - May 2023

University of Colorado Denver

- Directed development of a quick-release mechanism for enteral gastrostomy tubes, spanning mechanical design, prototyping, manufacturing, regulatory considerations, and validation testing.

SELECTED SYSTEMS

NWB Forge

Designed a PySide6 desktop workflow for converting heterogeneous neuroscience acquisition data into validated NWB files. Structured ingest, metadata normalization, mapping, assembly, provenance review, and recoverable validation steps.

Closed-Loop Reach Training and Acquisition Platform

Integrated synchronized video capture, DeepLabCut-enabled tracking, hardware control, and operator safeguards for long-running behavioral sessions. Supported deployment validation, maintenance, and daily operation.

EDUCATION

Bachelor of Science in Bioengineering
Graduated May 2023

University of Colorado Denver - Anschutz Medical Campus

- GPA: 3.7 | Dean's List, 2021-2023 | CEDC Design Expo Winner

CORE CAPABILITIES

Scientific software and data Python, NumPy, SciPy, Pandas, OpenCV, PySide6, NWB, PyNWB, NeuroConv	**Automation and acquisition** DeepLabCut, multi-camera capture, closed-loop control, frame synchronization, TTL timing, CAN bus
Deployment and systems Linux, Windows, Docker, Git, GitHub, Conda, CMake, PyInstaller, release builds, application deployment	**Hardware and product development** KiCad, Autodesk Inventor, AutoCAD, Fusion 360, SolidWorks, PCB layout, enclosures, prototyping, bench testing